

Economica Obscura: Institutions, Archives, and Policy Learning

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Institutions adapt installed reporting systems to current priorities and leave the resulting records to later users. The model combines operating and adjustment costs, rank-one reports, and policy loss from inherited precision. Under a fixed priority, reporting directions specialize geometrically. Precision orthogonal to the limiting direction remains finite, and a planner who values later use chooses slower specialization. State persistence governs how inherited precision depreciates. The archive also changes subsequent measurement: directional precision can rotate later reports, jointly chosen reports can straddle the operating priority, and report content updates beliefs about coefficients and future target regimes. The empirical application uses U.S. Call Reports. Form 051 persists at rates of 99.89 and 99.62 percent across adjacent observed quarters in two eligible cohorts. An out-of-time acquisition score reduces mortgage-composition MSE by 21.43, 25.90, and 23.44 percent under ridge, persistence, and gradient boosting during 2018–2025; forecast-specific rules select the same reports at least 95.58 percent of the time.

1. Introduction

Statistical records inherit the priorities under which institutions produce them. A supervisory report tracks exposures selected by a current mandate, while revisions to definitions, software, and reporting frequency use resources. Later officials inherit those records and may value directions that received limited attention during production.

This timing creates an intertemporal measurement problem. The operating institution pays current installation and reporting costs, while later policy users receive the precision stored in the archive. The analysis characterizes the reporting path chosen under current priorities, the policy loss borne by later users, and a payment that assigns archive value to the institution that produced it.

The sequence begins with institutional specialization and ends with the composition and later use of the archive. An organization chooses an economic response direction, and a statistical agency chooses a measurement direction. Each receives an operating return from following its priority and pays to change its installed system. Under a constant priority, quadratic costs generate the relative path $\delta_t - \bar{\delta} = \gamma^t(\delta_0 - \bar{\delta})$, where persistence γ increases with adjustment curvature. With rank-one Gaussian reports of a stable coefficient vector, precision along the limiting direction grows with the archive, while orthogonal precision converges to a finite value C_∞ . Proposition 1 compares the specialization speed chosen by the institution with the speed chosen by a planner who values the limiting policy loss.

State persistence determines how long inherited information remains useful. For $\theta_{t+1} = \phi\theta_t + \omega_{t+1}$, Theorem 2 shows that a record's value h dates later is ϕ^{2h} times its current value. Inherited precision also affects what the institution reports next. Theorem 4 gives the policy-loss weight at which the next report begins to rotate. Theorem 5 extends this threshold to a finite path, and Corollary 4 places two jointly chosen reports on opposite sides of a constant operating priority. In Theorem 3, report content updates both coefficient precision and the probability of a future target regime, so the realized signal changes the next reporting direction through a cutoff.

Target-weighted Bayesian A-optimality maps archive precision into a policy decision (Chaloner and Verdinelli 1995; Pukelsheim 2006). A policymaker choosing q to minimize $cq^2/2 - qa'\theta$ has expected posterior loss $a'J^{-1}a/(2c)$ relative to full information. Averaging across policy problems gives $\text{tr}(DJ^{-1})/2$, where $D = E[aa'/c]$. The planner combines this loss with operating and adjustment costs, and a path-contingent archive payment implements the planner allocation by assigning the policy-loss reduction to the institution that produced the reporting path.

U.S. Call Reports connect the model to observed reporting choices and downstream forecast loss. FFIEC 051 adoption changes the installed reporting schedule, agency burden estimates provide preparation-hour measures, and adjacent observed quarters retain form

051 at rates of 99.89 and 99.62 percent in two fixed eligible cohorts. Retained adjustable-rate mortgage reports define a pre-report state for acquiring an intermediate observation. An acquisition score estimated from earlier years targets the mean paired loss reduction across ridge, persistence, and gradient boosting. At a one-quarter item budget in 2018–2025, it reduces mortgage-composition MSE by 21.43, 25.90, and 23.44 percent. Forecast-specific acquisition rules select the same report as the common rule on at least 95.58 percent of evaluation origins.

Sections 3–5 develop institutional adjustment and the resulting archive. Section 6 studies reporting choices made from inherited precision and report content. Section 7 applies the framework to U.S. bank reporting. The online appendix contains proofs, posterior calculations, data construction, and predictive diagnostics.

2. Related Literature

Measurement choices respond to economic objectives and institutional incentives. Almås et al. (2024) develop measurement tools alongside decision models; Veldkamp and Chung (2024) survey data as an input to production and learning; and Liang and Madsen (2024) study how measurements change effort incentives. Work on multitask incentives, agency missions, sequential reporting, performance signals, and bureaucratic data collection includes Holmstrom and Milgrom (1991), Dewatripont et al. (1999), Li (2007), Ekmekci et al. (2022), and Oechslin and Steiner (2022). The model here assigns installation and reporting costs to operating institutions and archive value to subsequent policy users, producing the planner–institution wedge.

Parameter learning depends on cumulative information. Boyd and Sastry (1983) and Bai and Sastry (1985) link reference-input richness, persistent excitation, and parameter convergence; Ziemann and Sandberg (2021) characterize uninformative optimal policies through Fisher-information singularity. Lemma 3 states the cumulative-information condition used here, and Theorem 1 applies it to an installation path generated by adjustment costs.

Sequential experiment choice with switching costs appears in Chernoff (1959), Kiefer and Sacks (1963), Whittle (1965), Kolonko and Benzing (1985), and Hardwick and Stout (2007). Sensor-scheduling formulations combine estimation error, physical states, and switching costs (Baras and Bensoussan 1989; Lee et al. 2001; Leong et al. 2016). The reporting problem uses a continuous installed angle and rank-one precision updates. Theorems 4 and 5 characterize the onset of reorientation, while Corollary 4 and Theorem 6 characterize two-date paths.

The loss criterion $\text{tr}(DJ^{-1})/2$ is target-weighted Bayesian A-optimality (Chaloner and Verdinelli 1995; Pukelsheim 2006). Sequential Bayesian design coordinates belief and physical states (Huan et al. 2024), and belief cutoffs also arise in active-sensing control

(Ahmad et al. 2014). Dynamic information acquisition across signal processes appears in Zhong (2022) and Liang et al. (2022). Theorem 3 combines a coefficient posterior with a target-regime posterior in the next measurement choice.

Costly feature acquisition provides the empirical decision analogue. Saar-Tsechansky et al. (2009) rank feature acquisitions by estimated information value. The bank application ranks an intermediate report from its retained history and evaluates the ranking across forecasting procedures. Related policy-feedback studies include Lucas (1976), McLeay and Tenreyro (2020), and Chemla and Hennessy (2021); the appendix also uses results from majorization, matrix analysis, identification, and projection inference (Marshall et al. 2011; Horn and Johnson 2012; Escanciano 2022; Kaji 2021; Andrews and Mikusheva 2016; Andrews and Guggenberger 2019).

3. Institutions and the Observation Experiment

3.1. Response, Measurement, and Economic Units

Let $\theta \in \mathbb{R}^d$ collect economic coefficients with a fixed interpretation and unit for each coordinate. A local response map and an observation experiment have derivatives

$$(1) \quad G_r = V_r \Sigma W', \quad K_r = U_r \Lambda U_r', \quad \mathcal{J}_r = G_r' K_r G_r = W \Sigma R_r' \Lambda R_r \Sigma W', \quad R_r = U_r' V_r.$$

For the square specification, U_r, V_r, W are orthogonal, Σ contains response singular values, and Λ contains measurement-information eigenvalues. For example, $Y = H_r x + \varepsilon$, $x = G_r \theta$, and $\varepsilon \sim N(0, \Omega_r)$ give $K_r = H_r' \Omega_r^{-1} H_r$. The capacities Σ, Λ and the right basis W are held fixed across the comparisons that follow.

Fixed capacities give an isospectral structural experiment under the additional restriction $\Sigma = g I_d$, with $g > 0$. The analysis uses $W = I_d$ in the declared economic coordinates. A change of units transforms both the information matrix and the target gradient. The target cost below is invariant to that transformation. Spectral comparisons are made in the declared units, or relative to a declared positive-definite metric as described in Appendix A.

3.2. Operating Priorities

The static direction choices can be represented by value and exposure moments. Let $p, \zeta \in \mathbb{R}^d$ denote operating values and exposures, and let $M_r = E[p \zeta']$. An organization chooses V to maximize $E[p' V \Sigma W' \zeta] = \text{tr}(B_r' V)$, where $B_r = M_r W \Sigma$. For nonsingular B_r , its choice is

$$(2) \quad \bar{V}_r = B_r (B_r' B_r)^{-1/2}.$$

A statistical agency has monitoring values $C_r = E[\mathbf{v}\mathbf{v}']$ and chooses U to maximize $\text{tr}(C_r U \Lambda U')$. Its columns align ordered eigenvectors of C_r with the ordered eigenvalues of Λ . These operating problems determine desired directions conditional on the regime. Their derivations are in Appendix A.4.

The dynamic model below uses quadratic costs of deviations from those desired directions and of changing an installed direction. The quadratic operating costs are maintained primitives on an angular coordinate. They can represent local approximations to the operating criteria above; the dynamic results use them as the specified objective. Both decision makers observe the priority path. Their objectives are separable conditional on that path. The later policymaker benefits from the observations, and its loss is assigned to the planner's objective with weight χ . This allocation of objectives specifies the source of the information externality.

3.3. Two Directions

For $d = 2$, write

$$(3) \quad Q(x) = \begin{pmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{pmatrix}, \quad e(x) = \begin{pmatrix} \cos x \\ \sin x \end{pmatrix}, \quad V_t = Q(\alpha_t), \quad U_t = Q(\beta_t), \quad \delta_t = \beta_t - \alpha_t.$$

With $\Lambda = \text{diag}(\lambda_H, \lambda_L)$, where $\lambda_H > \lambda_L \geq 0$, one observation supplies

$$(4) \quad \mathcal{J}(\delta) = g^2 Q(\delta) \Lambda Q(\delta)' = g^2 \{ \lambda_L I_2 + (\lambda_H - \lambda_L) e(\delta) e(\delta)' \}.$$

An exact experiment yielding this matrix is $Y = g \Lambda^{1/2} Q(\delta)' \theta + \varepsilon$, with $\varepsilon \sim N(0, I_2)$. The second component carries zero signal when $\lambda_L = 0$. The signs of the bases are fixed by an installation convention. Angles in the adjustment problem are real-valued installed coordinates; observation precision is periodic with period π .

The information matrix in (4) describes a fresh observation. Retained records contribute to the archive used for the policy decision. Appendix A gives target costs, economic-unit transformations, and the Schur–Horn allocation set for fresh observations. The reporting paths and policy choices below use the accumulated posterior precision.

4. Intertemporal Institutional Adjustment

4.1. Objectives and Timing

At date t , an organization inherits α_t and chooses α_{t+1} . An agency inherits β_t and chooses β_{t+1} . The next report uses the new directions. Their desired positions are $\bar{\alpha}_t$ and $\bar{\beta}_t$, with $\bar{\delta}_t = \bar{\beta}_t - \bar{\alpha}_t$. For either decision maker, write the installed coordinate as x_t and its desired

coordinate as $\bar{x}_t$. The problem is

$$(5) \quad \min_{\{x_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \eta^t \left\{ \frac{\rho}{2} (x_{t+1} - \bar{x}_t)^2 + \frac{\kappa}{2} (x_{t+1} - x_t)^2 \right\}, \quad \rho, \kappa > 0, \quad 0 < \eta < 1.$$

The priority sequence is bounded and known at date zero. A constant sequence describes a regime expected to remain in force. Information about later changes can be incorporated by resolving the same problem after an announcement. Operating costs measure resources or losses that also enter the planner's objective. The parameters ρ, κ, η are common across the two institutions in the results using δ_t as their installed state.

4.2. Adjustment with Continuation Value

LEMMA 1 (Quadratic institutional adjustment). *Let $P > 0$ solve*

$$(6) \quad P = \frac{\kappa(\rho + \eta P)}{\rho + \kappa + \eta P}, \quad \gamma = \frac{\kappa}{\rho + \kappa + \eta P}.$$

For $0 < \eta < 1$, $P = \{\sqrt{[\rho + \kappa(1 - \eta)]^2 + 4\eta\kappa\rho} - [\rho + \kappa(1 - \eta)]\}/(2\eta)$. The unique optimal path satisfies

$$(7) \quad x_{t+1} = \gamma x_t + (1 - \gamma)\tilde{x}_t, \quad \tilde{x}_t = (1 - \eta\gamma) \sum_{j=0}^{\infty} (\eta\gamma)^j \bar{x}_{t+j}.$$

Consequently $\delta_{t+1} = \gamma\delta_t + (1 - \gamma)\tilde{\delta}_t$, where $\tilde{\delta}_t$ is the same weighted average of $\bar{\delta}_{t+j}$. For a constant priority $\bar{\delta}$, $\delta_t - \bar{\delta} = \gamma^t(\delta_0 - \bar{\delta})$. Holding ρ and η fixed, γ increases with κ .

Appendix B solves the Bellman equation and verifies the minimizing path. The continuation term ηP enters the denominator in (6). The case $\eta = 0$ gives the one-period coefficient $\kappa/(\rho + \kappa)$. For $\eta > 0$, anticipated priorities enter today's installation through the weights in (7).

An announced unit change confined to priority date $t + j$ changes x_{t+1} by $(1 - \gamma)(1 - \eta\gamma)(\eta\gamma)^j$, holding x_t fixed. A constant priority following a new announcement produces a geometric transition from the inherited position. These responses separate announcement timing, installation, and the date at which the resulting report enters the archive.

When operating or adjustment curvatures differ across institutions, each direction follows its own equation (7). The installed state is then (α_t, β_t) . With the common coefficients assumed above, the relative direction itself obeys the displayed scalar transition.

5. Data Histories and Policy Learning

5.1. The Archive and the Policy Decision

Assume a fixed parameter $\theta \sim N(\mu_0, J_0^{-1})$, where $J_0 > 0$. At reporting date s , n_s conditionally independent observations follow the experiment in (4). Their noises are independent across dates. The records include the direction, sample size, and measurement map for every date, and the policymaker retains all observations. The posterior precision after reports $0, \dots, T-1$ is

$$(8) \quad J_T = J_0 + \sum_{s=0}^{T-1} n_s \mathcal{J}(\delta_s).$$

The report at date zero can be inherited. The institutions subsequently choose the directions of reports $1, 2, \dots$ by (7). Reindexing the archive by one period leaves the results below unchanged.

After observing a policy problem (a, c) , with $c > 0$, the policymaker chooses $q \in \mathbb{R}$ to minimize $L(q, \theta; a, c) = cq^2/2 - qa'\theta$. Given θ , the minimizing action is $q^*(\theta) = a'\theta/c$. The distribution of (a, c) is independent of the parameter and observations, and $D = E[aa'/c]$ is finite. It may depend on a known policy regime.

LEMMA 2 (Bayesian quadratic decision loss). *Let μ_T be the posterior mean. The Bayes action is $\widehat{q} = a'\mu_T/c$, and its expected loss relative to $q^*(\theta)$ is*

$$(9) \quad \mathcal{R}(J_T; D) = E[L(\widehat{q}, \theta; a, c) - L(q^*(\theta), \theta; a, c) \mid J_T] = \frac{1}{2} \text{tr}(DJ_T^{-1}).$$

For fixed $\mathcal{J} > 0$, J_0 , and D , $\mathcal{R}(J_0 + n\mathcal{J}; D) = \text{tr}(D\mathcal{J}^{-1})/(2n) + O(n^{-2})$.

The equality uses a proper prior and the Gaussian experiment. In the expansion, $\mathcal{J}$ is fixed and positive definite. The target cost in (A1) describes the coefficient of $1/n$ under those conditions. The exact loss (9) applies throughout the Gaussian specification, including rank-one reports.

5.2. Adjustment Costs and the Information Left by Specialization

Suppose the priority remains at $\bar{\delta}$. Rotate coordinates so that $u = e(\bar{\delta})$ is the first coordinate and $v = e(\bar{\delta} + \pi/2)$ is the second. The rank-one case, $\lambda_L = 0$, makes each report informative about its measured linear combination. A finite set of nonparallel reports identifies both coefficients in the associated linear mean experiment. The rate at which new directional information accumulates determines posterior concentration.

LEMMA 3 (Cumulative excitation). *In the fixed-state Gaussian experiment with deterministic reporting maps and $J_0 > 0$, the posterior covariance converges to zero if and only if $\lambda_{\min}(\sum_{s < T} n_s \mathcal{I}(\delta_s)) \rightarrow \infty$. A uniform lower bound $\sum_{s=t}^{t+L-1} n_s \mathcal{I}(\delta_s) \geq \epsilon I_d$ for some fixed $L \geq 1$, $\epsilon > 0$, and every t implies covariance of order $O(T^{-1})$.*

The first condition concerns cumulative information in every direction. The second is a finite-window persistent-excitation condition of the type studied in adaptive control (Boyd and Sastry 1983; Bai and Sastry 1985). Sparse supplemental reports can satisfy the cumulative condition with increasingly long gaps between them. The proof uses eigenvalue bounds for J_T and appears in Appendix G.

THEOREM 1 (Adjustment costs and limiting archive precision). *Let $n_s = n > 0$, $\lambda_L = 0$, $\tau = ng^2\lambda_H$, and $\delta_s = \bar{\delta} + \gamma^s \epsilon_0$ with $0 < \gamma < 1$. Define*

$$(10) \quad C_\infty = v' J_0 v + \tau \sum_{s=0}^{\infty} \sin^2(\gamma^s \epsilon_0).$$

Then

$$(11) \quad J_T^{-1} \rightarrow \frac{vv'}{C_\infty}, \quad \mathcal{R}(J_T; D) \rightarrow \frac{v'Dv}{2C_\infty}, \quad C_\infty \leq v' J_0 v + \frac{\tau \epsilon_0^2}{1 - \gamma^2}.$$

For $0 < |\epsilon_0| \leq \pi/2$, C_∞ is strictly increasing in γ . For $v'Dv > 0$, the limiting policy loss therefore decreases with κ , holding the initial gap, priorities, ρ , η , and experiment fixed.

The path generated by adjustment contributes a finite amount of precision about the coefficient orthogonal to the eventual reporting direction. Increasing the adjustment cost lengthens this transition and preserves additional observations of that coefficient. This statement compares information along the decentralized paths. A welfare comparison also includes the operating and adjustment costs in (5).

The proof writes J_T in the (u, v) basis. Its first diagonal entry diverges linearly in T , its off-diagonal entry converges, and its second diagonal entry converges to C_∞ . Matrix inversion gives (11). The convergence of the off-diagonal entry uses the geometric rate as well as the boundedness of the angular gap.

To compare specialization speeds within the same technology, consider geometric paths $\delta_t - \bar{\delta} = \gamma^t \epsilon_0$ for $\gamma \in [0, 1]$. The discounted resource cost associated with the relative direction is

$$(12) \quad \mathcal{C}^G(\gamma) = \frac{\epsilon_0^2}{4} \frac{\rho \gamma^2 + \kappa(1 - \gamma)^2}{1 - \eta \gamma^2}.$$

For $\bar{d} = v'Dv > 0$, define $\mathcal{L}^G(\gamma) = \bar{d}/[2C_\infty(\gamma)]$ on $[0, 1]$ and $\mathcal{L}^G(1) = 0$. At $\gamma = 1$, the installed direction remains at its initial value and orthogonal precision diverges.

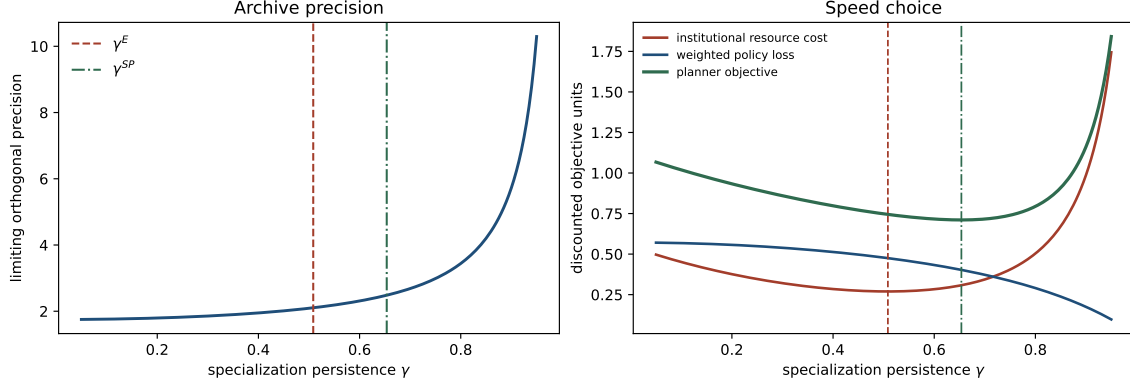


FIGURE 1. Specialization speed and the archive externality

The calculation uses $\rho = 1$, $\kappa = 2$, $\eta = 0.95$, $\varepsilon_0 = \pi/3$, $v'J_0v = \tau = \bar{d} = 1$, and $\chi = 2$. The decentralized choice is $\gamma^E = 0.508$ and the planner's choice is $\gamma^{SP} = 0.654$. The left panel evaluates (10); the right panel evaluates (12), weighted limiting policy loss, and their sum.

PROPOSITION 1 (The planner's specialization speed). *Let $0 < |\varepsilon_0| \leq \pi/2$, $\rho, \kappa > 0$, $\eta \in (0, 1)$, and $\bar{d} > 0$. The decentralized speed γ^E uniquely minimizes $\mathcal{C}^G$ and satisfies*

$$(13) \quad \rho\gamma^E = \kappa(1 - \gamma^E)(1 - \eta\gamma^E).$$

For every $\chi > 0$, each global minimizer of $\mathcal{C}^G(\gamma) + \chi\mathcal{L}^G(\gamma)$ lies in $(\gamma^E, 1]$. For all sufficiently small $\chi > 0$, this minimizer is unique and interior, and

$$(14) \quad \gamma^{SP} = \gamma^E + \chi \frac{\bar{d}\mathcal{C}'_\infty(\gamma^E)}{2\mathcal{C}_\infty(\gamma^E)^2\mathcal{C}^{G''}(\gamma^E)} + O(\chi^2) > \gamma^E.$$

The operating institution chooses γ^E from current operating and reconfiguration costs. The planner values the precision accumulated during the transition and delays specialization. Equation (14) measures the local speed wedge in policy demand, archive sensitivity, and the curvature of resource cost.

COROLLARY 1 (Measurements in the remaining direction). *If $\lambda_L > 0$ and $n_s = n$, then $\mathcal{R}(J_T; D) \leq \text{tr}(D)/(2Tng^2\lambda_L)$ for $T \geq 1$. Under the rank-one path of Theorem 1, supplement the archive with m_T independent reports of $v'\theta$, each with precision $\xi > 0$. If $m_T \rightarrow \infty$, the posterior covariance tends to zero. This conclusion also holds when $m_T/T \rightarrow 0$.*

The two statements give observation requirements for posterior concentration. A positive per-report precision floor supplies information in every direction at every date. Supplemental reports supply a growing total precision in the direction left by the convergent installation path.

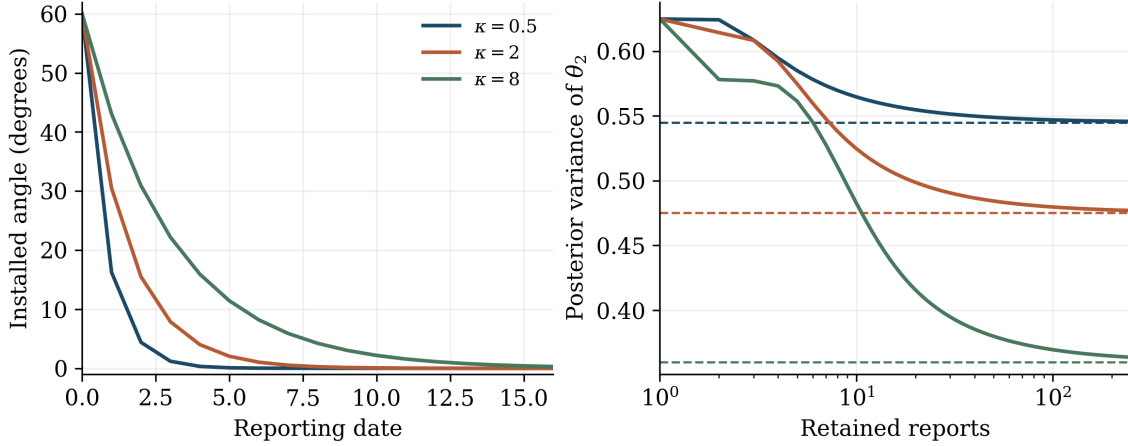


FIGURE 2. Installation speed and posterior uncertainty

Computed examples use $\rho = 1$, $\eta = 0.95$, $\kappa \in \{0.5, 2, 8\}$, $J_0 = I_2$, $\tau = 1$, $\bar{\delta} = 0$, and $\varepsilon_0 = \pi/3$. The left panel plots the optimal installed direction. The right panel plots posterior variance of the second coefficient; dashed lines are the limits in Theorem 1. Values are generated by the Gaussian model.

5.3. Changes in the Parameter or the Archive

Equation (8) uses a stable coefficient vector and retained records with known observation maps. If the coefficient follows $\theta_{t+1} = \theta_t + \omega_{t+1}$, with independent Gaussian innovation covariance $Q_t \geq 0$, prediction and updating give

$$(15) \quad J_{t+1}^- = (J_t^{-1} + Q_t)^{-1}, \quad J_{t+1} = J_{t+1}^- + n_{t+1} J(\delta_{t+1}).$$

The decision loss remains (9) for the current posterior. Theorem 1 uses $Q_t = 0$. A retained subset of records contributes its own sum of information matrices. Loss of regime labels changes the likelihood itself; Appendix F states the information difference between labelled and unlabelled observations. These specifications make parameter drift, record retention, and map retention explicit inputs to the learning problem.

5.4. An Additional Record and a Future Target

The archive can also be represented by its posterior covariance P_t . For a Gaussian state with transition $\theta_{t+1} = A\theta_t + \omega_{t+1}$ and independent innovation covariance Q , prediction gives $P_{t+1}^- = AP_t A' + Q$. A report $r = s'\theta_t + \varepsilon$, with independent variance $\nu \geq 0$, updates this covariance to $P_t - P_t s s' P_t / (\nu + s' P_t s)$ whenever the denominator is positive. These are the covariance recursions of Kalman (1960). They allow a current coordinate to be observed exactly while leaving uncertainty about another coordinate, such as its subsequent rate of change.

LEMMA 4 (Archive-conditional value of a record). *Suppose the report and the subsequent state innovations satisfy the Gaussian specification above, and acquiring the report leaves the state transition unchanged. For a policy decision h dates later with loss matrix $D \geq 0$, the reduction in expected posterior decision loss is*

$$(16) \quad \Delta_h(P_t; s, v, D) = \frac{s' P_t (A^h)' D A^h P_t s}{2(v + s' P_t s)}.$$

The value is zero exactly when $D^{1/2} A^h P_t s = 0$. For a scalar forecast target $u' \theta_{t+h}$ under squared error, its expected MSE reduction is (16) with $D = uu'$ and the factor $1/2$ removed.

The numerator measures how uncertainty in the report covaries with uncertainty in the future target after propagation through A^h . The denominator measures the report's predictive variance. Thus two histories that imply different P_t can give different values to the same report at the same current installation. A target whose residual uncertainty has little covariance with the report has little expected gain. The application constructs such an uncertainty state, evaluates its forecast calibration, and compares the predicted gain with losses observed after an additional record becomes available.

5.5. State Persistence and Effective Archive Age

The fixed-state archive in Theorem 1 retains every precision contribution. A changing state discounts earlier records through the transition matrix in Lemma 4. The following result separates the value of record age from the steady filtering loss created by a specialized reporting system.

THEOREM 2 (State persistence and a specialized archive). *Suppose $\theta_{t+1} = \phi \theta_t + \omega_{t+1}$, where $|\phi| < 1$ and $\omega_{t+1} \sim N(0, qI_2)$ with $q > 0$. First, for any report described in Lemma 4,*

$$(17) \quad \Delta_h(P_t; s, v, D) = \phi^{2h} \Delta_0(P_t; s, v, D), \quad \sum_{h=0}^H \eta^h \Delta_h = \Delta_0 \frac{1 - (\eta \phi^2)^{H+1}}{1 - \eta \phi^2}.$$

Second, suppose that from date $T + 1$ onward every report has rank-one precision $\tau e_1 e_1'$, with $\tau > 0$, and that $P_T = \text{diag}(p_{1,T}, p_{2,T})$. The posterior covariance remains diagonal and converges to $P^ = \text{diag}(p_1^*, p_2^*)$, where*

$$(18) \quad p_2^* = \frac{q}{1 - \phi^2}, \quad p_1^* = \begin{cases} \frac{\sqrt{(1 + \tau q - \phi^2)^2 + 4\tau\phi^2 q} - (1 + \tau q - \phi^2)}{2\tau\phi^2}, & \phi \neq 0, \\ \frac{q}{1 + \tau q}, & \phi = 0. \end{cases}$$

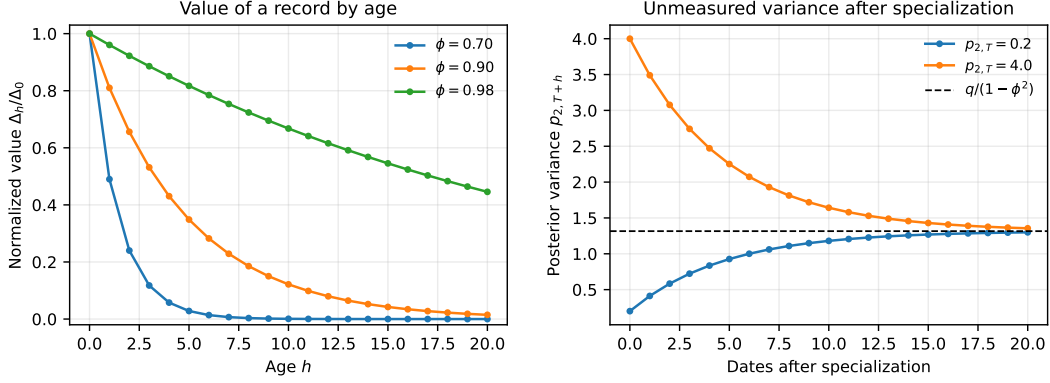


FIGURE 3. State persistence and the effective age of a record

The left panel plots the normalized value $\Delta_h/\Delta_0 = \phi^{2h}$. The right panel plots the unmeasured-coordinate variance after specialization for $q = 0.25$, $\phi = 0.9$, and two inherited covariances. Its limit is $q/(1 - \phi^2)$. Values follow Theorem 2.

For two diagonal covariances P_T and $\tilde{P}_T$ followed by the same specialized reports,

$$(19) \quad |p_{1,T+h} - \tilde{p}_{1,T+h}| \leq |\phi|^{2h} |p_{1,T} - \tilde{p}_{1,T}|, \quad |p_{2,T+h} - \tilde{p}_{2,T+h}| = |\phi|^{2h} |p_{2,T} - \tilde{p}_{2,T}|.$$

Consequently, for $D \geq 0$, the posterior policy-loss difference obeys

$$(20) \quad |\mathcal{R}(P_{T+h}; D) - \mathcal{R}(\tilde{P}_{T+h}; D)| \leq \frac{|\phi|^{2h}}{2} \{D_{11}|p_{1,T} - \tilde{p}_{1,T}| + D_{22}|p_{2,T} - \tilde{p}_{2,T}|\}.$$

Equation (17) gives a record an economic half-life determined by state persistence. Equation (18) gives the current-state uncertainty produced by permanent specialization: the measured coordinate reaches a scalar filtering variance, while the other coordinate reaches its unconditional stationary variance. Reports collected during installation change finite-horizon losses according to (19). Their effect lasts longer when $|\phi|$ is closer to one.

Theorems 1 and 2 describe two archive regimes. A stable coefficient preserves the directional precision collected during institutional transition. A stationary changing state depreciates that precision and makes recent directional coverage enter current policy loss. Adjustment cost then affects the timing of information benefits, while state persistence determines how much of each benefit reaches later users. This distinction matches the application, where removing recent reports changes composition forecasts more than adding older records.

6. The Planner's Reporting Choice

6.1. A State Consisting of Installation and Precision

The planner includes the costs of both institutions and assigns weight $\chi \geq 0$ to policy losses. Write $m = (\alpha + \beta)/2$. Their combined one-period cost separates into a component for the common direction and

$$(21) \quad C_t(x, y) = \frac{\rho}{4}(y - \bar{\delta}_t)^2 + \frac{\kappa}{4}(y - x)^2, \quad x = \delta_t, \quad y = \delta_{t+1}.$$

The common direction follows its own quadratic problem. Information depends on y , so its contribution can be removed from the planner's relative-direction problem. For dates $t = 0, \dots, H-1$, let the known sequence n_t describe the next reporting batches and let D_t describe the policy problems following those batches. With $\mathcal{V}_H = 0$, the planner's value is

$$(22) \quad \mathcal{V}_t(x, J) = \min_{y \in \mathbb{R}} \left\{ C_t(x, y) + \frac{\chi}{2} \text{tr} \left(D_t [J + n_t \mathcal{J}(y)]^{-1} \right) + \eta \mathcal{V}_{t+1}(y, J + n_t \mathcal{J}(y)) \right\}.$$

In (22), J is the precision available before the next reporting batch. When target demand is fixed and Gaussian posterior covariance does not depend on report realizations, the planner's state is therefore (x, J) together with the date and exogenous schedule. The posterior mean determines the policy action and integrates out of expected regret. The objective allocates institutional costs and the benefits received by later policy users, with χ expressing that allocation in the units of operating cost.

Conditional on the initial archive, a sequence of chosen angles determines the entire precision path in the baseline Gaussian specification. A date-zero reporting schedule then implements the planner allocation, while realized report values determine subsequent policy actions through the posterior mean. Section 6.2 uses the first report to update both the Gaussian coefficient posterior and a latent target-regime posterior. The recursion there has state (x, μ, P, π) , and the realized report changes (μ, π) while its precision changes P . Two states with the same installation and coefficient precision can therefore induce different next reports.

For finite H , positive J , and finite D_t , an optimal choice exists. The quadratic adjustment term makes the objective coercive in y . Moreover, $\mathcal{V}_t(x, J)$ decreases in the Loewner order on J . At a differentiable branch of the value function, define $J^+ = J + n_t \mathcal{J}(y)$ and

$$(23) \quad M_t = \frac{\chi}{2} (J^+)^{-1} D_t (J^+)^{-1} - \eta \nabla_J \mathcal{V}_{t+1}(y, J^+).$$

The matrix M_t is positive semidefinite. An interior optimum satisfies

$$(24) \quad \frac{\rho}{2}(y - \bar{\delta}_t) + \frac{\kappa}{2}(y - x) + \eta \partial_x \mathcal{V}_{t+1}(y, J^+) = n_t \text{tr}\{M_t \mathcal{J}'(y)\}.$$

The right side is the marginal value of changing the next batch's information. It includes the effects on the current policy decision and on subsequent uses of the archive. Appendix D derives this condition and its differentiability qualification.

6.2. One Report and Two Posterior States

The baseline state (x, J) records installation and precision when future target demand is known. A report can also reveal which policy problem will use the archive. Let $\theta \sim N(0, S)$ with $S = \text{diag}(s_1, s_2)$, and let a latent target index satisfy $T = \alpha + \lambda\theta_1 + \zeta$, where $\zeta \sim N(0, \sigma_T^2)$ is independent and $\lambda > 0$. The target regime is $Z = 1\{T \geq 0\}$. Conditional on $Z = 1$, future policy demand assigns weight d_H to the first coordinate and d_L to the second; the weights are reversed under $Z = 0$, where $d_H > d_L \geq 0$.

The installed system first reports

$$(25) \quad Y = \theta_1 + \epsilon, \quad \epsilon \sim N(0, r), \quad r > 0.$$

Write $k = s_1/(s_1 + r)$, $p_1 = s_1 r/(s_1 + r)$, and $\sigma_*^2 = \sigma_T^2 + \lambda^2 p_1$. Gaussian conditioning gives

$$(26) \quad \begin{aligned} \mu(y) &= E(\theta \mid Y = y) = (ky, 0)', \quad P = \text{Var}(\theta \mid Y = y) = \text{diag}(p_1, s_2), \\ \pi(y) &= \Pr(Z = 1 \mid Y = y) = \Phi\left(\frac{\alpha + \lambda ky}{\sigma_*}\right). \end{aligned}$$

Thus the first report changes coefficient precision from S^{-1} to $J = P^{-1}$ and its realized content changes conditional target demand,

$$(27) \quad D\{\pi(y)\} = \text{diag}\{d_L + (d_H - d_L)\pi(y), d_H - (d_H - d_L)\pi(y)\}.$$

The next report has precision $\tau > 0$ and is placed at e_1 or e_2 . A move from the installed coordinate e_1 to e_2 costs $c > 0$. Put $A(p) = \tau p^2/(1 + \tau p)$. Conditional on $Y = y$, the expected posterior policy loss after placing the next report at e_i is

$$(28) \quad R_i(y) = \frac{1}{2} \text{tr} \left[D\{\pi(y)\} \left\{ P - \frac{\tau P e_i e_i' P}{1 + \tau e_i' P e_i} \right\} \right].$$

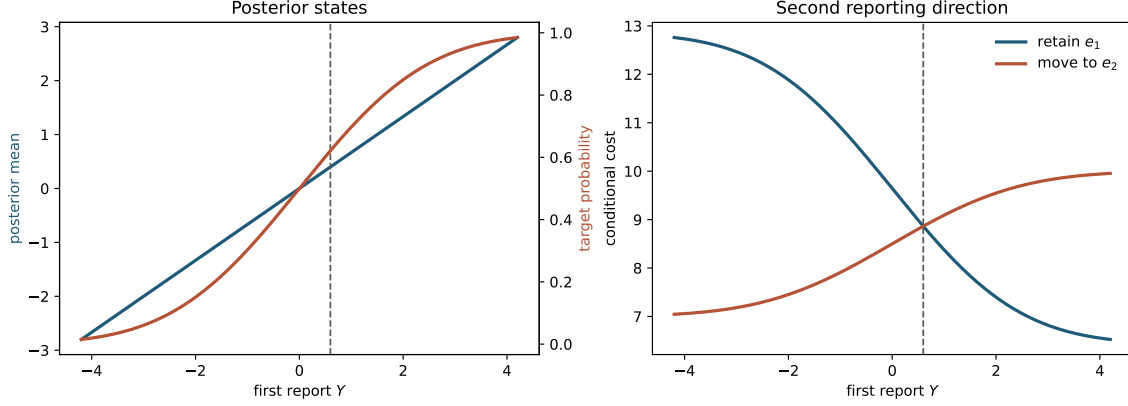


FIGURE 4. One report updates the archive and target demand

The left panel maps the first report realization into the posterior coefficient mean and the probability of target regime one. Coefficient precision changes from S^{-1} to P^{-1} for every realization. The right panel plots conditional costs for the two possible second reports; their crossing gives $y^\dagger$. Parameters are $S = \text{diag}(2, 1)$, $r = 1$, $\alpha = 0$, $\lambda = 1$, $\sigma_T^2 = 1$, $\tau = 2$, $d_H = 4$, $d_L = 1$, $\chi = 6$, and $c = 1$.

THEOREM 3 (A report-content cutoff after a precision update). *Define $A_1 = A(p_1)$, $A_2 = A(s_2)$, $\Delta_d = d_H - d_L$, and*

$$(29) \quad \pi^\dagger = \frac{d_H A_2 - d_L A_1 - 2c/\chi}{\Delta_d(A_1 + A_2)}.$$

For $\chi > 0$, the planner moves the next report to e_2 exactly when $\pi(Y) < \pi^\dagger$ and retains e_1 exactly when $\pi(Y) > \pi^\dagger$. If $\pi^\dagger \in (0, 1)$, the report rule has the realization cutoff

$$(30) \quad Y < y^\dagger, \quad y^\dagger = \frac{\sigma_* \Phi^{-1}(\pi^\dagger) - \alpha}{\lambda k}.$$

Let $v(y) = \min\{\chi R_1(y), c + \chi R_2(y)\}$. A plan that selects the second direction after observing the first report satisfies

$$(31) \quad E\{v(Y)\} \leq \min[E\{\chi R_1(Y)\}, E\{c + \chi R_2(Y)\}].$$

The inequality is strict when Y reaches both action regions with positive probability.

The first observation supplies both arguments of the second reporting decision. Its measurement precision determines P , and its realization determines $D\{\pi(Y)\}$. The cutoff compares target-weighted variance reductions $d_1\{\pi(y)\}A_1$ and $d_2\{\pi(y)\}A_2$ with the movement cost. A larger inherited variance along a coordinate raises the variance reduction available from measuring that coordinate. A realization associated with target regime one raises the weight on the first coordinate and shifts the second report toward e_1 . The content state extends the planner recursion to (x, μ, P, π) . After a report realization, Bayes'

rule maps this state into (μ', P', π') , and future directions can depend on all four entries. Equation (31) prices the option to choose the next report after observing the first one. The timing uses the report once: it enters the coefficient posterior used in policy loss and the target posterior used in policy demand.

6.3. Institutional Costs and Benefits to Archive Users

For the finite-horizon institutional problem, let C_H be the discounted sum of the two institutions' resource costs after removing the common direction, and let R_H be the discounted sum of posterior policy losses. The decentralized path δ^E minimizes C_H ; the planner's path δ^{SP} minimizes $C_H + \chi R_H$. Their objective difference is

$$(32) \quad \mathcal{W}_H = \chi \{R_H(\delta^E) - R_H(\delta^{SP})\} - \{C_H(\delta^{SP}) - C_H(\delta^E)\} \geq 0.$$

Current institutions bear the second term, and subsequent users receive the first. The inequality follows by including δ^E in the planner's feasible set. It measures the welfare wedge in the units specified by χ and the resource costs. Under the constant-priority assumptions of Theorem 5, $\delta^E = 0$ and $\mathcal{W}_H > 0$ precisely above its threshold. A comparison across values of κ changes the underlying resource technology as well as reporting behavior; equation (32) compares paths at a fixed technology.

PROPOSITION 2 (An archive-value payment). *Suppose an archive user can commit before reporting to a path-contingent payment*

$$(33) \quad T_H(\delta) = \chi \{R_H(\delta^E) - R_H(\delta)\}.$$

An institution choosing a feasible path to minimize $C_H(\delta) - T_H(\delta)$ selects the planner's path. The payment is zero at δ^E . At every planner solution distinct from δ^E , it is positive and the archive user's gross decision-loss reduction equals the payment. The net social gain equals (32) because the payment is a transfer.

PROOF. The institution's net cost is $C_H(\delta) + \chi R_H(\delta) - \chi R_H(\delta^E)$, so its minimizers equal the planner's minimizers. Strict convexity of C_H gives $C_H(\delta^{SP}) > C_H(\delta^E)$ whenever $\delta^{SP} \neq \delta^E$. Planner optimality then implies

$$(34) \quad T_H(\delta^{SP}) \geq C_H(\delta^{SP}) - C_H(\delta^E) > 0.$$

Subtracting the resource-cost increase from this payment gives (32). □

The payment uses quantities already present in the planner's problem: declared policy targets, inherited precision, reporting maps, and institutional resource costs. A reporting standard can implement the same allocation by restricting the feasible path to a planner

solution. In the payment formulation, the information externality is the reduction in policy loss created for later users.

6.4. Inherited Precision and the Next Report

The terminal reporting choice yields a solved case of (22). Set $x = \bar{\delta} = 0$, $h = \rho + \kappa$, and let inherited precision be $J = \text{diag}(j_1, j_2)$, with $j_1 \geq j_2 > 0$. A batch of n reports contributes

$$(35) \quad n\mathcal{J}(\delta) = c_0 I_2 + \tau e(\delta)e(\delta)', \quad c_0 = ng^2\lambda_L, \quad \tau = ng^2(\lambda_H - \lambda_L) > 0.$$

Use $D = I_2$, which can be generated by policy target $a = \sqrt{2}e_1$ or $\sqrt{2}e_2$ with equal probability and $c = 1$. Define $a_0 = j_1 + c_0$, $b_0 = j_2 + c_0$, and

$$(36) \quad K = b_0(a_0 + \tau), \quad L = \tau(a_0 - b_0), \quad N = a_0 + b_0 + \tau.$$

The planner minimizes

$$(37) \quad F(\delta) = \frac{h}{4}\delta^2 + \frac{\chi}{2} \frac{N}{K + L \sin^2 \delta}.$$

For any information-equivalent angle, the planner selects a representative with $|\delta| \leq \pi/2$ because it has the smallest installation displacement from zero.

THEOREM 4 (Archive precision and reporting direction). *If $a_0 = b_0$, the unique minimizer of (37) is zero. If $a_0 > b_0$, let*

$$(38) \quad \chi^* = \frac{hK^2}{2NL} = \frac{hb_0^2(a_0 + \tau)^2}{2\tau(a_0 - b_0)(a_0 + b_0 + \tau)}.$$

For $0 \leq \chi \leq \chi^$, the unique global minimizer is zero. For $\chi > \chi^*$, the global minimizers are $\pm\delta^*$, where $\delta^* \in (0, \pi/2)$ is the unique positive solution of*

$$(39) \quad h\delta^* = \frac{\chi NL \sin(2\delta^*)}{[K + L \sin^2 \delta^*]^2}.$$

The magnitude δ^ increases with χ/h , tends to zero as $\chi \downarrow \chi^*$, and tends to $\pi/2$ as $\chi \rightarrow \infty$. Holding b_0, τ, h fixed, χ^* strictly decreases with a_0 , from $+\infty$ as $a_0 \downarrow b_0$ to $hb_0^2/(2\tau)$ as $a_0 \rightarrow \infty$.*

Both signs rotate toward the other coordinate. Their equality reflects the reflection symmetry of the specified archive, policy demand, and operating target. The threshold in (38) is a global statement. Its proof uses the exact objective difference

$$(40) \quad F(\delta) - F(0) = \frac{h}{4}\delta^2 - \frac{\chi NL \sin^2 \delta}{2K(K + L \sin^2 \delta)}.$$

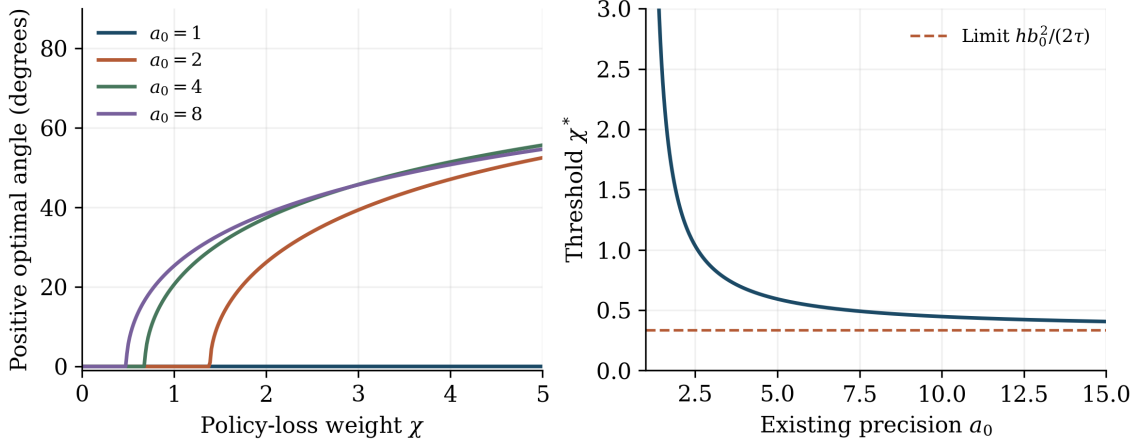


FIGURE 5. The next reporting angle conditional on the archive

Computed examples use $b_0 = 1$, $\tau = 3$, and $h = 2$. The left panel plots the positive optimal angle against χ for $a_0 \in \{1, 2, 4, 8\}$. The right panel plots the threshold in (38); the dashed line is $hb_0^2/(2\tau)$. A negative optimum of equal magnitude exists above each threshold.

The inequality $\sin^2 \delta \leq \delta^2$ verifies optimality of zero below the threshold. Above it, the curvature at zero is negative and the positive stationary equation has one solution. Appendix D.3 provides the uniqueness argument and the derivative of the threshold.

COROLLARY 2 (Operating continuation after the policy decision). *Suppose the planner chooses one report, policy loss is incurred after that report, and the institutions subsequently solve (5) with constant zero priorities. Starting from $\alpha = \beta = 0$, the reporting choice is characterized by Theorem 4 with h replaced by $h_c = \rho + \kappa + \eta P$, where P solves (6).*

The continuation term prices the subsequent operating and adjustment costs caused by the intervention. It follows from summing the two institutions' value functions and removing their common direction, as shown in Appendix D.4.

The archive creates the directional policy incentive in this example. When $j_1 = j_2$, isotropic demand values every rotation of the new batch equally. With $j_1 > j_2$, a rotation toward the second coordinate increases the determinant of posterior precision while preserving its trace. The inverse trace in (37) then falls. Additional historical precision about the first coefficient lowers the policy-loss weight at which paying for the rotation becomes worthwhile. The adjustment cost remains $h\delta^2/4$ throughout this comparison.

6.5. A Local Reporting Change and Its Value

For a general terminal state, completing the square in (21) gives $h(\delta - \delta^E)^2/4$ plus a constant, where $\delta^E = (\rho\bar{\delta} + \kappa x)/h$. Define $r(\delta) = \text{tr}\{D[J + n\mathcal{J}(\delta)]^{-1}\}/2$. For a branch of minimizers

converging to δ^E as $\chi \rightarrow 0$, the implicit function theorem gives

$$(41) \quad \delta^{SP} - \delta^E = -\frac{2\chi}{h} r'(\delta^E) + O(\chi^2), \quad F(\delta^E) - F(\delta^{SP}) = \frac{\chi^2}{h} [r'(\delta^E)]^2 + O(\chi^3).$$

Here $r'(\delta) = -n \operatorname{tr}\{J_\delta^{-1} D J_\delta^{-1} \mathcal{J}'(\delta)\}/2$, with $J_\delta = J + n\mathcal{J}(\delta)$. The local change depends on the precision already available and on policy demand. At the symmetric state in Theorem 4, $r'(0) = 0$. The global comparison in that theorem supplies the positive threshold and the subsequent reporting choices.

6.6. Reporting Paths and the Archive They Create

Consider H reporting dates, a stable coefficient vector, a constant zero priority, $\delta_0 = 0$, $D_t = I_2$, and rank-one batches with precision $\tau e(\delta_t) e(\delta_t)'$, where $\tau > 0$. Let the initial archive be $J_0 = \operatorname{diag}(a, b)$ with $a \geq b > 0$. The objective in (22) becomes

$$(42) \quad F_H(\delta) = \frac{1}{4} \sum_{t=1}^H \eta^{t-1} \{ \rho \delta_t^2 + \kappa (\delta_t - \delta_{t-1})^2 \} + \frac{\chi}{2} \sum_{t=1}^H \eta^{t-1} \operatorname{tr}(J_t^{-1}), \quad J_t = J_0 + \tau \sum_{s=1}^t e(\delta_s) e(\delta_s)'. \quad J_t = J_0 + \tau \sum_{s=1}^t e(\delta_s) e(\delta_s)'.$$

Here $\rho, \kappa > 0$ and $0 < \eta < 1$. The report at each date changes the archive used for every remaining policy decision. Writing $S_t = a + b + t\tau$ and $K_t = b(a + t\tau)$, the determinant is

$$(43) \quad \det J_t = K_t + \tau(a - b) \sum_{s=1}^t \sin^2 \delta_s + \tau^2 \sum_{1 \leq r < s \leq t} \sin^2(\delta_r - \delta_s).$$

The last sum records information produced by differences between reporting dates. It enters even when the initial archive has $a = b$.

Define symmetric $H \times H$ matrices A_H and B_H through their quadratic forms, with $z_0 = 0$:

$$(44) \quad \begin{aligned} z' A_H z &= \sum_{t=1}^H \eta^{t-1} \{ \rho z_t^2 + \kappa (z_t - z_{t-1})^2 \}, \\ z' B_H z &= \sum_{t=1}^H \frac{\eta^{t-1} S_t}{K_t^2} \left\{ \tau(a - b) \sum_{s=1}^t z_s^2 + \tau^2 \sum_{1 \leq r < s \leq t} (z_r - z_s)^2 \right\}. \end{aligned}$$

THEOREM 5 (A threshold for a sequence of reporting choices). *The objective (42) has a minimizer. If $B_H \neq 0$, define*

$$(45) \quad \chi_H^* = \frac{1}{2\lambda_{\max}(A_H^{-1/2} B_H A_H^{-1/2})}.$$

For $0 \leq \chi \leq \chi_H^$, the unique minimizing path is $\delta_t = 0$ at every date. For $\chi > \chi_H^*$, every minimizing path has a nonzero angle. If $B_H = 0$, the zero path is uniquely optimal at every finite χ . With*

$a = b$, $B_1 = 0$ and $B_H \neq 0$ for $H \geq 2$. With $H = 1$ and $a > b$, (45) equals the threshold in Theorem 4 for $\lambda_L = 0$.

The threshold compares the operating and adjustment costs of an entire path with its effects on all subsequent archives. The proof uses (43) to bound the objective over every path; the generalized eigenvalue then determines where the zero path first loses optimality. The two-date case also yields every minimizing path.

COROLLARY 3 (Reporting costs and the path threshold). *Fix H, a, b, τ, η with $B_H \neq 0$. The threshold χ_H^* is strictly increasing in each of ρ and κ . Scaling both costs by $c > 0$ scales the threshold by c .*

These comparisons keep information production fixed. The discounted difference operator with $z_0 = 0$ and the operating-cost matrix are positive definite, so each cost increase raises the Rayleigh quotient defining the threshold. Appendix H.6 gives the argument. For two dates, Corollary 4 also signs the discount-factor comparison. Figure A1 displays longer-horizon calculations for a stated parameter grid.

COROLLARY 4 (Two dates with equal initial precision). *Let $H = 2$ and $a = b = j > 0$. Define $h = \rho + \kappa$, $K = j(j + 2\tau)$, $S = 2j + 2\tau$, and*

$$(46) \quad c = \kappa + \frac{\rho h}{h + \eta \rho}, \quad \chi_2^* = \frac{cK^2}{2S\tau^2}.$$

For $\chi \leq \chi_2^$, both reporting angles are zero. For $\chi > \chi_2^*$, there are two minimizing paths, related by reflection. The path with positive separation $z = \delta_2 - \delta_1$ satisfies*

$$(47) \quad \delta_1 = -\frac{\eta \rho}{h + \eta \rho} z, \quad \delta_2 = \frac{h}{h + \eta \rho} z, \quad cz = \frac{\chi S \tau^2 \sin 2z}{(K + \tau^2 \sin^2 z)^2}, \quad 0 < z < \frac{\pi}{2}.$$

There is one such positive z . It increases with χ/c , tends to zero as χ decreases to χ_2^ , and tends to $\pi/2$ as χ increases without bound. The threshold increases with κ and decreases with η .*

The first report by itself has the same expected policy loss at every angle because the initial archive and demand are isotropic. Its chosen angle anticipates the second report. Above the threshold, the institution first moves to one side of its operating priority and then moves to the other side. Its first observation creates an archive with unequal precision, and its second observation uses the less measured direction. The operating priority and policy demand stay constant throughout this sequence. The result characterizes two dates; equation (45) gives the onset of reorientation for any finite horizon.

To examine the role of the isotropic benchmark, let $F_2(\delta; \zeta)$ denote the two-date objective with $J_0 = \text{diag}(a, b)$, $D = \text{diag}(d_1, d_2)$, first- and second-report precisions (τ_1, τ_2) , and the operating parameters in (42). The vector ζ collects these positive quantities. Let ζ_0

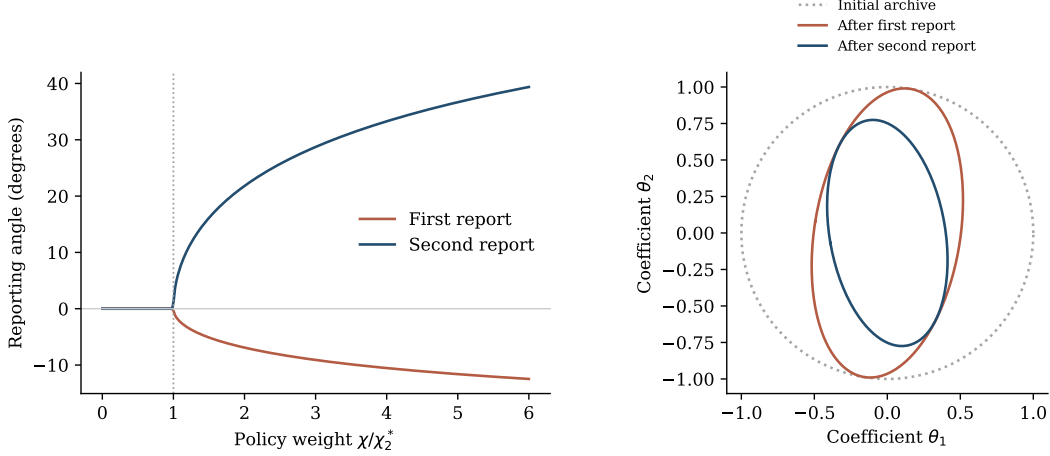


FIGURE 6. Jointly chosen reports under a constant priority

The calculations use $j = 1$, $\tau = 3$, $\rho = 1$, $\kappa = 2$, and $\eta = 0.95$. The left panel plots the two minimizing angles on the branch with positive separation against χ/χ_2^* . The right panel traces the posterior covariance ellipses after the first and second reports at $\chi = 3\chi_2^*$. These are model calculations. The reflection of the displayed path has the same cost.

set $a = b = j$, $d_1 = d_2 = 1$, and $\tau_1 = \tau_2 = \tau$ while retaining the other parameters in Corollary 4.

THEOREM 6 (Straddling under nearby precision and target demand). *Fix $\chi > \chi_2^*$ at ζ_0 , and let $\delta^0 = (\delta_1^0, \delta_2^0)$ be the minimizing path with $\delta_1^0 < 0 < \delta_2^0$. There are a neighborhood $\mathcal{U}$ of ζ_0 , two continuously differentiable functions $\delta^+(\zeta)$ and $\delta^-(\zeta)$, and a finite constant C such that every global minimizer of $F_2(\cdot; \zeta)$ for $\zeta \in \mathcal{U}$ equals one of these two branches and*

$$(48) \quad \min_{q \in \{\delta^0, -\delta^0\}} \|\delta^s(\zeta) - q\| \leq C \|\zeta - \zeta_0\|, \quad s \in \{+, -\}.$$

Each branch places its two reports on opposite sides of the zero operating priority: $\delta_1^s(\zeta)\delta_2^s(\zeta) < 0$.

The theorem uses the positive curvature at the two benchmark minima and the objective gap separating them from the remaining paths. Precision and target-demand asymmetry can therefore change the two angles while preserving their ordering around the operating priority. The neighborhood depends on the benchmark distance above χ_2^* and the angular margins in (47).

7. Archives in U.S. Bank Reporting

7.1. The Reporting Institution and the Targets

U.S. Call Reports record mortgage composition under different reporting schedules. In March 2017, eligible domestic institutions could adopt form FFIEC 051, which generally

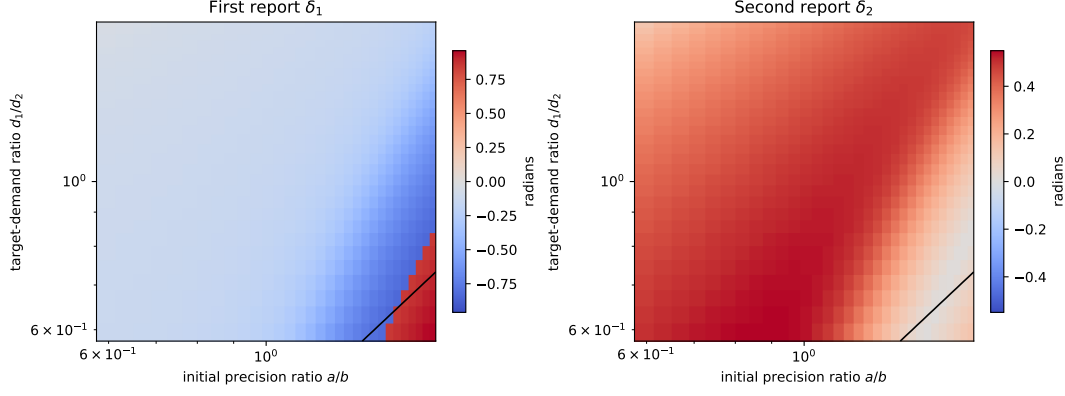


FIGURE 7. Two-date reporting with asymmetric precision and demand

The calculations set $j = 1$, $\tau_1 = \tau_2 = 3$, $\rho = 1$, $\kappa = 2$, $\eta = 0.95$, and $\chi = 3\chi_2^*$. The horizontal axis changes initial precision through a/b while preserving $ab = 1$; the vertical axis changes policy demand through d_1/d_2 while preserving $d_1d_2 = 1$. Each grid point uses 25 starting values for the two-angle global search. Both reports straddle zero throughout the rectangle $a/b, d_1/d_2 \in [0.8, 1.25]$.

used an asset threshold of one billion dollars at introduction (Office of the Comptroller of the Currency, Federal Reserve Board, and Federal Deposit Insurance Corporation 2017). Its adjustable-rate first-lien mortgage item, RCON5370, is reported in June and December. The mortgage total and the six maturity or repricing buckets remain available quarterly (Federal Financial Institutions Examination Council 2016). The application changes access to the dated ARM item while retaining the other observed records. The reporting form and item schedule describe the observed installation; retained ARM reports define the archive; subsequent mortgage composition and net interest income define two forecast targets.

The source contains 315,615 bank-quarter observations over 2012–2025, downloaded from the FDIC BankFind API on August 28, 2026 (Federal Deposit Insurance Corporation 2026). Reporting metadata, historical identifiers, mergers, and quarterly financial variables accompany the records. Federal funds and two- and ten-year Treasury rates come from FRED (Federal Reserve Bank of St. Louis 2026). The sample consists of domestic commercial banks on forms 041 or 051 with positive assets and first-lien mortgages, complete common controls, and observed outcomes. Histories restart at mergers, identifier changes, and gaps. The accounting and continuing-bank conditions in Appendix M yield 165,049 evaluated bank-origin observations for 2016–2025. Calculations use the downloaded vintage. An origin means that quarter’s reports are available; historical release times and first-release values are outside the recorded data.

Write A_{it} for ARM balances, T_{it} for first-lien mortgage totals, and S_{it} for bank assets. Put $h_t = 1$ for March and September origins and $h_t = 2$ for June and December origins. The

outcomes are

$$(49) \quad Y_{it}^A = 100 \frac{A_{i,t+h_t}}{T_{i,t+h_t}}, \quad Y_{it}^N = 10^4 \frac{\text{INTINQ}_{i,t+1} - \text{EINTXQ}_{i,t+1}}{S_{it}}.$$

Composition is measured in percentage points and income in basis points of origin assets. Income inputs are quarterly flows. Composition targets fall on June or December reports. March and September form-051 ARM entries follow the reporting mask, including positive entries found in the downloaded file. Each access comparison holds the bank, origin, other records, and realized outcome fixed.

7.2. Reform Timing and Observed Form Choices

The 2017 introduction removed about 40 percent of the items in form 041 and changed reporting frequencies for selected items (Office of the Comptroller of the Currency, Federal Reserve Board, and Federal Deposit Insurance Corporation 2017). Further revisions effective in June 2018 affected about seven percent of the streamlined form's items (Federal Financial Institutions Examination Council 2018). In September 2019, eligibility expanded to certain domestic institutions with assets below five billion dollars, and approximately 37 percent of first- and third-quarter items were removed from those quarters (Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation, and Office of the Comptroller of the Currency 2019). The ARM item retained its June/December schedule for continuing 051 filers. Form adoption by newly eligible banks is therefore the channel connecting this expansion to their ARM archives.

Figure 8 follows fixed cohorts defined in the preceding June. Among 4,654 domestic commercial banks below one billion dollars in June 2016, 3,015 use form 051 at the first effective filing and 344 first use it later through December 2018. Among 455 banks with June 2018 assets from one billion to below five billion dollars, 81 use 051 in September 2019 and 40 first use it later through December 2020. The plotted denominator is the cohort present within the domestic-commercial-bank form sample at each date. These paths document implementation and choice following the reporting rule. Table 1 follows adjacent observed form choices within the same fixed cohorts. Once a bank reports on 051, the next observed quarter retains that form in 99.89 percent of 2017-cohort transitions and 99.62 percent of 2019-cohort transitions. Entry from 041 occurs in 4.18 and 2.59 percent of adjacent transitions. The observed installation therefore changes through an adoption margin followed by persistence in the selected form. Adoption timing reflects reporting savings, system changes, expected growth, and supervisory requirements. The 2019 rule estimates average quarterly preparation time of 40.11 to 39.08 hours for existing 051 filers and 63.69 to 50.96 hours for newly eligible filers (Board of Governors of the Federal Reserve System, Federal Deposit Insurance Corporation, and Office of the Comptroller of the

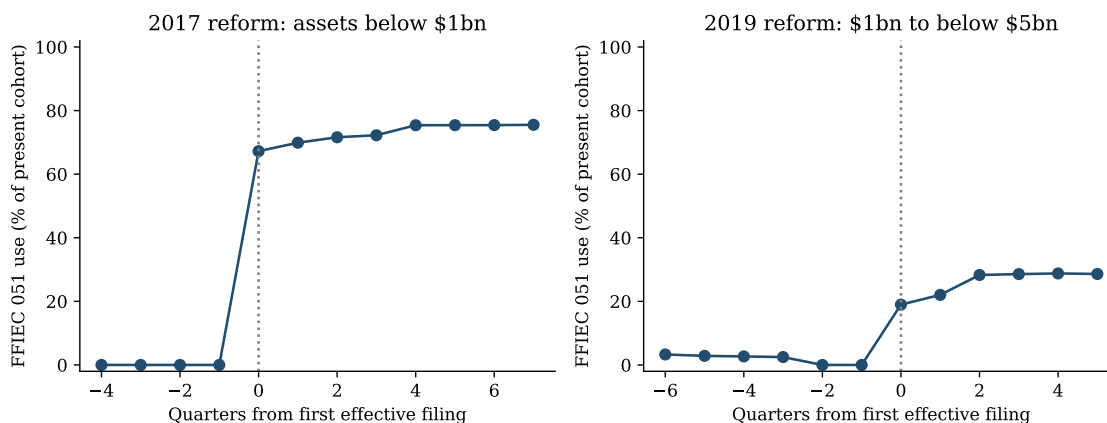


FIGURE 8. Form use after eligibility changes

Cohorts use June 2016 assets for the 2017 reform and June 2018 assets for the 2019 reform. The first effective filings are March 2017 and September 2019. Presence requires the domestic commercial-bank and form conditions described in Appendix Q. The vertical line marks the first effective filing. The plotted denominator follows banks present in the cohort sample at each date.

TABLE 1. Adoption timing and persistence of the installed reporting form

Eligible cohort	Banks	Adopt	Immediate	Retain 051	Enter 051
2017: below \$1bn	4,654	74.89	89.76	99.89	4.18
2019: \$1-5bn	455	28.34	66.94	99.62	2.59

Percentages use fixed pre-reform asset cohorts. Any adoption divides banks that use FFIEC 051 by banks observed after the effective date. Immediate adoption is the share of adopters first observed on FFIEC 051 at the effective filing. Retention and entry use adjacent observed quarters through 2018Q4 for the 2017 cohort and 2020Q4 for the 2019 cohort. The transition rates describe installed-form histories; bank choice and eligibility conditions determine those histories.

Currency 2019). The 2017 notice records a commenter's estimate of 160 conversion hours, primarily for training, and \$350 for software; the agencies cite heterogeneity in systems, automation, training, and vendor charges and leave conversion costs unquantified at the agency level (Office of the Comptroller of the Currency, Federal Reserve Board, and Federal Deposit Insurance Corporation 2017). Table 2 combines these figures at a stated labor price. The resulting payback differs by more than a factor of twelve across filer groups. These whole-form figures anchor the flow and adjustment costs associated with reporting changes. Item-level ARM costs and shared preparation costs enter the acquisition decision separately.

The FFIEC began its 2027 full review of Call Reports in October 2025 and collected industry comments on reporting burden and data use through January 2026. Its 2025 annual report places any resulting Paperwork Reduction Act notices by the end of 2026 (Federal Financial Institutions Examination Council 2026). This review preserves the

TABLE 2. FFIEC reporting-burden accounting

Filer group	Before	After	Hours saved	Annual dollars	Payback quarters
Existing 051 filers	40.11	39.08	1.03	206	162.1
Newly eligible filers	63.69	50.96	12.73	2,546	13.1

Before, after, and hours saved are average preparation hours per quarterly filing reported in the 2019 final rule. Annual dollars value preparation time at \$50 per hour. Payback combines a commenter’s estimate of 160 conversion hours and \$350 of software cost with the estimated quarterly time saving. The agencies cited variation in training, systems, automation, and vendor costs and left conversion costs unquantified at the agency level. The entries provide an accounting scale for the operating and adjustment costs in the model.

policy setting in which reporting content, filer resources, and subsequent data use are considered together.

The local first stage is small in 2017 and zero in the 2019 window. Within ten percent of the 2017 threshold, nine of 59 present below-threshold banks use 051 in March, compared with zero of 58 above. Within ten percent of the 2019 threshold, 16 below and ten above are present in September; both sides have zero adopters, also in March 2020. Appendix Q records the eligibility and contemporaneous-rule qualifications. The application uses these histories as institutional evidence about reporting schedules and estimates access values on fixed realized outcomes.

7.3. A Posterior State from Retained Reports

The empirical state describes a changing mortgage share under a Gaussian-increment approximation. Over a retained window, let the increment between reports separated by d quarters satisfy $y_{i,s+d} - y_{is} = db_i + \epsilon_{is,d}$, where $\epsilon_{is,d} \mid q_i \sim N(0, dq_i)$ and disjoint increments are independent. The drift prior is $b_i \mid q_i \sim N(0, q_i/\lambda_0)$, and q_i has an inverse-gamma prior. The oldest retained report anchors the level. Reports observe composition exactly; uncertainty about its subsequent movement depends on the retained increments. The two-coordinate state consists of current composition and its drift, with innovation variance learned from the same history.

For archive H_{it} , let d_{it} be the age of its latest report, $\widehat{b}_{it} = E[b_i \mid H_{it}]$, $\widehat{q}_{it} = E[q_i \mid H_{it}]$, and $\lambda_{it} = \lambda_0 + \text{span}(H_{it})$. The predictive mean, variance, and inverse posterior drift variance are

$$\begin{aligned}
 \widehat{m}_{it}(h) &= y_{i,\text{last}} + (d_{it} + h)\widehat{b}_{it}, \\
 V_{it}(h) &= \widehat{q}_{it} \left\{ d_{it} + h + \frac{(d_{it} + h)^2}{\lambda_{it}} \right\}, \quad j_{it}^b = \frac{\lambda_{it}}{\widehat{q}_{it}}.
 \end{aligned}
 \tag{50}$$

At a current report, $d_{it} = 0$ and the observed level coincides across retained histories. Those histories can still imply different drift uncertainty and innovation variance. With q_i

TABLE 3. Retained histories with the same current report

Period	Records	Origins	RMSE	Mean V	MSE gain (%)
2016	4	18,070	6.381	21.04	0.00
2016	8	18,070	6.361	31.67	0.62
2016	12	18,070	6.369	37.02	0.36
2022–2025	4	33,482	7.608	27.35	0.00
2022–2025	8	33,482	7.608	40.94	0.01
2022–2025	12	33,482	7.611	48.26	-0.07
2016–2025	4	101,921	7.821	27.07	0.00
2016–2025	8	101,921	7.807	39.63	0.35
2016–2025	12	101,921	7.804	45.18	0.43

Every row within a period uses the same bank-origin observations and current ARM report. RMSE is in percentage points and variance in squared percentage points. MSE gains use four records as the reference. Posterior innovation variance is learned separately from each retained history.

fixed, a longer span reduces the drift contribution to future variance. When q_i is learned, older records also revise the estimated amount of variation. The full posterior includes that variance state. Appendix K derives its Student- t predictive distribution.

Prior tuning parameters are selected by predictive log score on the preceding year’s outcomes and held fixed for the target year. Bank-specific posteriors use retained records dated through the origin. Appendix K.3 gives the annual selection rule and predictive diagnostics.

7.4. Holding the Current Report Fixed

There are 101,921 bank-origin observations with a current ARM report and twelve available records. Supplying four, eight, or twelve records keeps the current report and all other current bank records identical within each comparison. Table 3 reports future forecast losses and posterior variances. Over 2016–2025, increasing the archive from four to twelve reports changes RMSE from 7.821 to 7.804 percentage points, a 0.43 percent MSE reduction. The two-way bank and target-date interval for the paired MSE reduction is $[-0.12, 0.65]$ squared percentage points. Average predictive variance rises as longer histories revise estimated innovation variance. The uncertainty state is evaluated against subsequent errors. Within each origin, banks are grouped by current composition decile and asset quintile. Removing the means of these joint cells, the slope of realized squared error on $V_{it}(h_t)$ is 0.381, with a two-way interval of $[0.297, 0.464]$. Table 4 reports the first year, 2022–2025, and the pooled sample. The association uses future errors and an archive state calculated at the origin. The paired retention comparison fixes the current observation exactly; the cross-bank calibration uses the stated bins.

TABLE 4. Archive states and subsequent forecast losses

Prediction	Period	Origins	Slope	Bank 95% CI	Two-way 95% CI
Uncertainty	2016	18,070	0.334	[0.161, 0.507]	[−0.182, 0.849]
Uncertainty	2022–2025	33,482	0.448	[0.315, 0.581]	[0.288, 0.608]
Uncertainty	2016–2025	101,921	0.381	[0.304, 0.457]	[0.297, 0.464]
Report gain	2016	9,799	0.402	[0.136, 0.667]	—
Report gain	2022–2025	7,085	0.338	[0.060, 0.617]	[−0.075, 0.751]
Report gain	2016–2025	29,588	0.374	[0.222, 0.526]	[0.276, 0.472]

Uncertainty regresses subsequent squared error on predicted variance after removing joint origin, current-composition-decile, and asset-quintile cell means. Report gain regresses realized paired MSE reduction on its ex ante prediction, with cells using latest semiannual composition. Two-way intervals cluster by bank and ARM target date. The 2016 sample has two target dates. A dash records a negative estimated two-way variance. Slopes condition on fitted prior parameters.

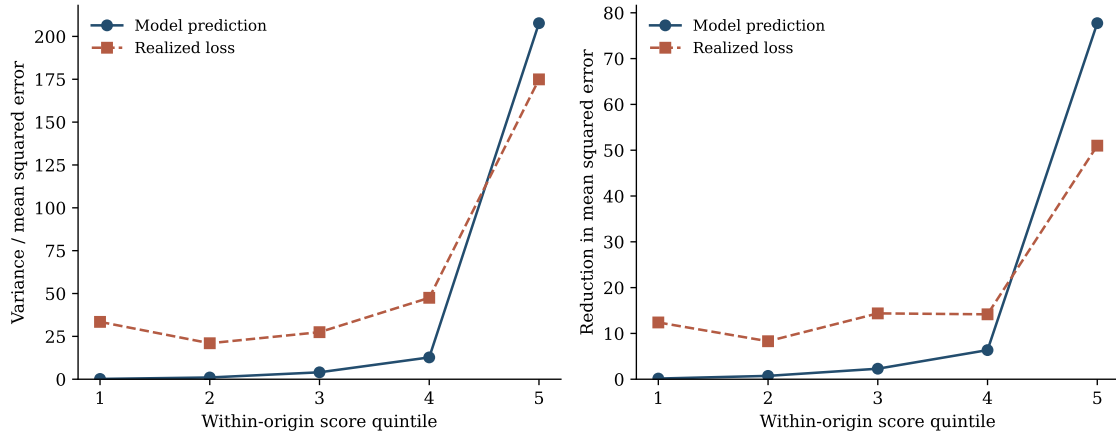


FIGURE 9. Archive uncertainty and the value of an additional record

The left panel uses 101,921 current-report origins with twelve retained observations during 2016–2025. Quintiles use model predictive variance within each origin; the lines compare average variance with subsequent squared error. The right panel uses 29,588 observed March and September origins. Quintiles use expected MSE reduction before the current report is supplied; the lines compare it with realized paired loss reduction. Units are squared percentage points. Forecasts and scores use the annual out-of-time fitting rule.

Figure 9 shows the scale of the calibration error. In the lowest within-origin variance quintile, average predicted variance is 0.17 and realized MSE is 33.46 squared percentage points. In the highest quintile, the corresponding values are 207.72 and 174.93. Large composition changes occur among banks with quiet retained histories. These outcomes qualify the absolute uncertainty estimates even where the archive state predicts variation in subsequent losses.

TABLE 5. Predictive calibration with repeated reported values

Model	RMSE	MSE/V	Cov.	Quiet MSE/V	Quiet cov.
Expanded continuous	8.692	1.09	87.0	748.90	91.3
Local-scale t	8.592	1.03	94.0	3.07	98.3
Point-mass/change	8.664	0.97	95.5	1.04	97.4

Coverage columns report percent coverage of 90 percent predictive intervals; V denotes mean predictive variance. Quiet histories are the lowest fifth of baseline semiannual predictive variance in the 2017–25 out-of-time sample. The mixture assigns an atom to the latest retained value and a Student t_5 distribution to changes. Each target year uses earlier target years to estimate repeat probabilities and twenty conditional-variance cells.

7.5. Prior Profiles and Predictive Calibration

Extending the prior grid toward smaller innovation scales places the continuous-density criterion at its lower boundary in every validation year. Conditional profiles separate constant retained histories from histories with changes, and Appendix P.2 gives the boundary calculation. A point-mass/change likelihood then assigns probability to repeated reported values and a continuous distribution to changes.

Across alternative variance specifications, the quiet-history group remains the main source of absolute calibration error. The point-mass/change model addresses the repeated-value component directly and is evaluated out of time. The online appendix reports the alternative variance fits and interval coverage. The sequential mixture uses 2016 as its first training year and predicts each target year from 2017 through 2025 using earlier target years. Exact repetition of the latest retained reported share occurs in 19.50 percent of the 145,569 out-of-time observations; the mean predicted probability is 19.46 percent. Annual repeat classifiers have AUC between 0.993 and 0.997. Relative to an expanding-window constant repeat probability with the same change distribution, the fitted probability lowers mixed negative log score by 0.446 per observation. Table 5 reports predictive calibration. The mixture’s MSE-to-variance ratio is 0.97 overall and 1.04 in the quiet-history fifth, compared with 3.07 for the local-scale benchmark. Its central intervals cover 95.5 percent overall and 97.4 percent in that fifth; probability mass at a reported value can make central-interval coverage exceed its nominal level. Appendix P.3 gives the likelihood, features, chronology, and decile diagnostics. Across 0, 1, 5, and 10 basis-point resolution bands, the observed low-change share rises from 19.50 to 25.63 percent, the mean fitted probability follows within 0.18 percentage point, and the predictive MSE-to-variance ratio stays between 0.970 and 0.972. Exact equality of the target ARM balance occurs in 19.01 percent of origins, while equality of both the ARM balance and mortgage-total balance occurs in 0.06 percent. The repeated-share component therefore largely reflects an unchanged ARM balance with a changing mortgage denominator. Appendix P gives the parameter grids, transformations,

moments, and fitting rules. The next calculation evaluates acquisition rankings using separate predictors and realized losses.

7.6. Valuing a Report Before Its Contents Are Supplied

Start from up to twelve June and December reports and consider acquiring the current March or September report. Every baseline record is retained, so an archive with twelve entries grows to thirteen. The Gaussian-increment model gives the ex ante score

$$(51) \quad g_{it}^G = \widehat{q}_{it} \left\{ d_{it} + h_t + \frac{(d_{it} + h_t)^2}{\lambda_{it}} - h_t - \frac{h_t^2}{\lambda_{it} + d_{it}} \right\}.$$

The score uses the semiannual archive and the known horizon before the current report value is supplied. Its drift-uncertainty term accounts for 6.26 percent of total fitted gain, and innovation scale accounts for most of its ranking. The mixed predictive state adds the probability that the intermediate report repeats the latest share, the conditional mean and variance of a change, the implied report variance, and g_{it}^G to the pre-report archive and current controls.

For forecast procedure $m \in \{R, P, B\}$, let $\widehat{y}_{it,S}^m$ use the semiannual archive and let $\widehat{y}_{it,S+}^m$ also use the intermediate report. Define its realized paired gain and the acquisition target by

$$(52) \quad G_{it}^m = (Y_{it}^A - \widehat{y}_{it,S}^m)^2 - (Y_{it}^A - \widehat{y}_{it,S+}^m)^2, \quad \bar{G}_{it} = \frac{1}{3} \sum_{m \in \{R, P, B\}} G_{it}^m.$$

The three procedures are ridge, persistence, and gradient boosting. A histogram boosting regression estimates $g^M(H_{it}) = E(\bar{G}_{it} | H_{it})$. Regularization is selected by five-fold bank validation in target year 2017 and fixed thereafter. For each evaluation year, the regression is refitted using earlier target years, and its score is applied before the intermediate report. The out-of-time allocation period is 2018–2025.

At each origin the acquisition rule selects the largest $\lfloor 0.25n_t \rfloor$ values of $g^M(H_{it})$. It uses one score across the three forecast procedures. Table 6 evaluates that selection separately for each procedure. Over 16,370 out-of-time bank origins, the rule reduces pooled MSE by 21.43, 25.90, and 23.44 percent for ridge, persistence, and gradient boosting. Uniform acquisition gives 9.93, 11.72, and 10.97 percent, and innovation-scale selection gives 20.18, 24.13, and 22.60 percent. Relative to uniform acquisition, the gains per eligible item are 5.482, 6.354, and 5.599 squared percentage points, and each two-way 95 percent interval lies above zero. Relative to innovation-scale selection, the point differences are 0.594, 0.795, and 0.380, with intervals that contain zero. During 2022–2025, the corresponding MSE reductions are 24.24, 28.75, and 25.26 percent. Three regressions trained separately on G_{it}^R , G_{it}^P , and G_{it}^B choose the same reports as the common rule on at least 95.58 percent

TABLE 6. A common mixed-state acquisition rule across forecast models

Forecast model	Uniform	Innovation scale	Mixed value
Ridge	9.93	20.18	21.43
Persistence	11.72	24.13	25.90
Gradient boosting	10.97	22.60	23.44

Forecast model	Δ vs. uniform	Δ vs. innovation
Ridge	5.482 [3.780, 7.185]	0.594 [-1.264, 2.452]
Persistence	6.354 [4.297, 8.411]	0.795 [-1.294, 2.884]
Gradient boosting	5.599 [3.957, 7.241]	0.380 [-1.396, 2.155]

The upper panel reports percentage reductions in pooled MSE relative to semiannual access during 2018–2025. The lower panel reports gain per eligible item in squared percentage points with 95 percent intervals clustered by bank and target quarter. Each origin acquires one quarter of observed intermediate reports. One acquisition score estimates the equal-weight average paired loss reduction across ridge, persistence, and gradient boosting from earlier target years. Its inputs contain the point-mass/change state of the intermediate report, the inherited archive, and current controls.

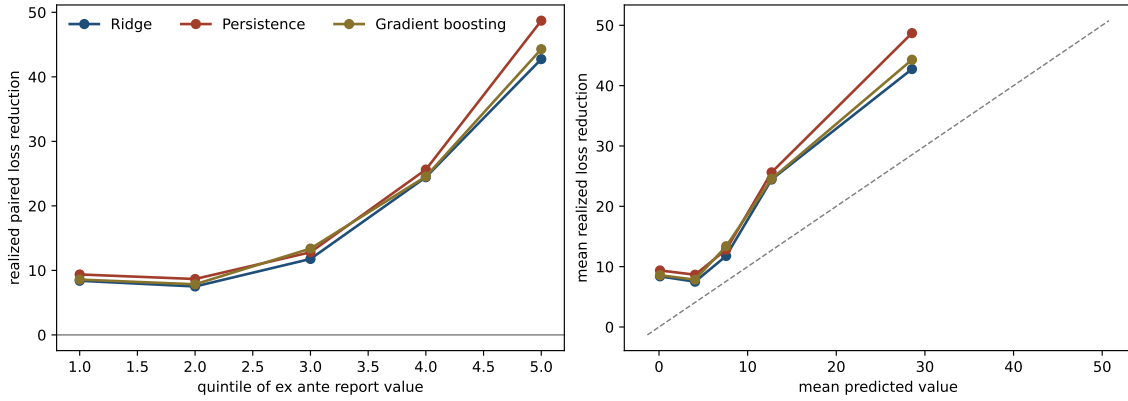


FIGURE 10. Ex ante report value and realized paired loss reduction

The score estimates the equal-weight average paired gain across ridge, persistence, and gradient boosting using earlier target years. The left panel orders observations into within-year fifths of the score. The right panel plots mean predicted value against mean realized gain for each forecast procedure. Units are squared percentage points.

of evaluation origins. Their common-minus-specific value differences are 0.125, 1.538, and -0.097 squared percentage points; the first and third clustered intervals contain zero, while the second lies above zero. Appendix Table A12 reports the comparison. Figure 10 groups the ex ante score into within-year fifths. Realized paired gains rise across its range for each forecast procedure. The acquisition exercise evaluates a common downstream target under three mappings from records to predictions; variation in G_{it}^m supplies the target-specific decision loss. With a monetary conversion ω_D dollars per squared percentage point of downstream loss and item cost c_{it} , the net acquisition value

is $\omega_{DG^M}(H_{it}) - c_{it}$. An unconstrained planner acquires the report when this expression is nonnegative; a budgeted planner solves the associated binary knapsack problem. The Call Report data identify forecast-loss gains and report counts. The FFIEC burden figures in Table 2 measure whole-form hours. Item-level time, shared preparation costs, and ω_D determine a monetary acquisition policy.

The Gaussian archive score, assets, predictive variance, and point-mass/change plug-in value appear in Appendix L. Those comparisons trace the source of the ranking and show how the common mixed-state score differs from a single variance component.

7.7. Income Information and the Downstream Target

The 2016 schedule comparison gives a 17.02 percent composition MSE reduction and a 0.15 percent next-quarter income MSE reduction. Removing current earnings from the ridge predictor changes the marginal ARM gain to 0.600 percent in 2016; in observed March/September origins during 2022–2025, the corresponding estimate is -0.106 percent. The full-information estimates for those samples are 0.146 and 0.065 percent. The paired intervals for changes in the marginal gain include zero. These results describe target-specific value conditional on the predictor and its existing records. The 2016 income persistence RMSE is 5.761 basis points, compared with 10.770 for ridge. Appendix R retains the earnings/maturity ablations, interval coverage, and fitting details.

7.8. The Scope of the Empirical Correspondence

The reform sequence supplies observed changes in reporting frequency, form adoption paths, adjacent-quarter form persistence, estimated quarterly preparation hours, and a stated conversion-cost benchmark. These objects correspond to flow reporting costs, installed form choice, and reconfiguration resources. The threshold windows contain too few adopters for a reform-based causal estimate of adjustment curvature, so the burden accounting provides a scale while the curvature remains a model parameter.

The archive posterior, its propagation to future composition, and the expected value of an acquired report instantiate Lemma 4. The record-length and recency comparisons correspond to Theorem 2: the value of an old record depends on the state transition linking it to the target date. The mixed-state rule then maps the inherited history and current controls into an out-of-time estimate of the report’s downstream loss reduction. Its one-quarter allocation is evaluated under three forecast procedures and against uniform, innovation-scale, predictive-variance, asset, and plug-in rules.

The data identify report schedules, form choices, whole-form preparation hours, archive-conditioned forecast-loss gains, and the content of acquired reports. Mapping these quantities into the planner’s monetary objective requires item-level costs, policy

actions, and the conversion ω_D . Desired angular directions and response maps also remain model primitives. The reported accounting and break-even formula keep each unmeasured quantity visible in the welfare calculation.

8. Conclusion

Institutional adjustment determines which directions are measured during a transition. With rank-one reports of stable coefficients, geometric specialization leaves only finite precision orthogonal to the eventual priority. Over the angular range of Theorem 1, a higher adjustment cost preserves more information in that direction. When the speed of specialization is itself a choice, Proposition 1 shows that a planner who values later decisions chooses a higher persistence parameter than the operating institution. If the underlying state changes over time, the value of an inherited record decays at a rate governed by state persistence, and permanent specialization returns the unmeasured coordinate to its stationary uncertainty.

The archive also changes future reporting incentives. Greater inherited precision along the installed direction lowers the threshold for reorientation. Joint planning values the effect of each report on the archives available at later dates. With two dates, the chosen reports lie on opposite sides of a constant priority, and this ordering persists under nearby asymmetries in precision and target demand. The first report may also update both coefficient beliefs and the posterior distribution of the future policy target. Its realized value then selects the next reporting direction through a probability cutoff. An archive-value payment assigns the resulting reduction in decision loss to the institution that produces the records.

The bank application connects changes in reporting forms to published estimates of preparation hours and conversion costs. In both fixed eligible cohorts, FFIEC 051 retention exceeds 99.62 percent across adjacent observed quarters. Retained histories then determine the value of acquiring another report. A mixed state of repeated values and changes remains calibrated across four reporting-resolution bands. A common acquisition score trained on earlier years improves ridge, persistence, and gradient-boosting forecasts under one held-out selection. Its selected reports overlap with forecast-specific rules at least 95.58 percent of the time. These estimates measure archive value under the declared forecast losses. A staggered form-eligibility design with verified reporting costs could identify form-assignment effects and turn the planner wedge into a policy experiment. Composition and income targets assign different values to the same ARM record. A monetary welfare calculation also needs the costs of individual report items and a conversion from forecast loss; the paper reports the corresponding break-even rule and keeps these inputs separate from the forecast-loss estimates.

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